ProductionManagementAI — Product master — Product Type — Database Design Document (テーブル定義書)

`004_DB-TYPE` defines the database side of 004_BD-TYPE (WI-017, REQ-090–REQ-093). It is additive: 004_DB and 004_DB-EXISTING-SCREENS stay unedited (RFC 0015), and everything not mentioned here is as they specify. PostgreSQL 17, `snake_case`, as 004_DB.

Document control (改版履歴)

Field	Value
Document ID	004_DB-TYPE
Work item	WI-017
Database / schema	`production_management_ai` / `public`
DBMS / encoding	PostgreSQL 17 / UTF-8
Created / updated	Claude (for ThanhTN), 2026-10-25

Version	Date	Author	Revision content
1	2026-10-25	Claude (for ThanhTN)	Initial creation: `products.product_type` (WI-017 DEC-001, DEC-002, DEC-004, DEC-005)

No.	Target	Change
1	`products`	Add required `product_type` with an allow-list CHECK
2	`products` data	Backfill: per-SKU mapping for the 30 demo parts, `FINISHED` for any other row
3	`pmai_app` privileges	Add `product_type` to the column-scoped `UPDATE` grant

Table list (テーブル一覧)

No.	Schema	Physical name	Logical name	Status
1	`public`	`products`	Product master	Existing; one column added

No other table changes. `production_orders` and every order read stay as they are (DEC-003).

ER diagram

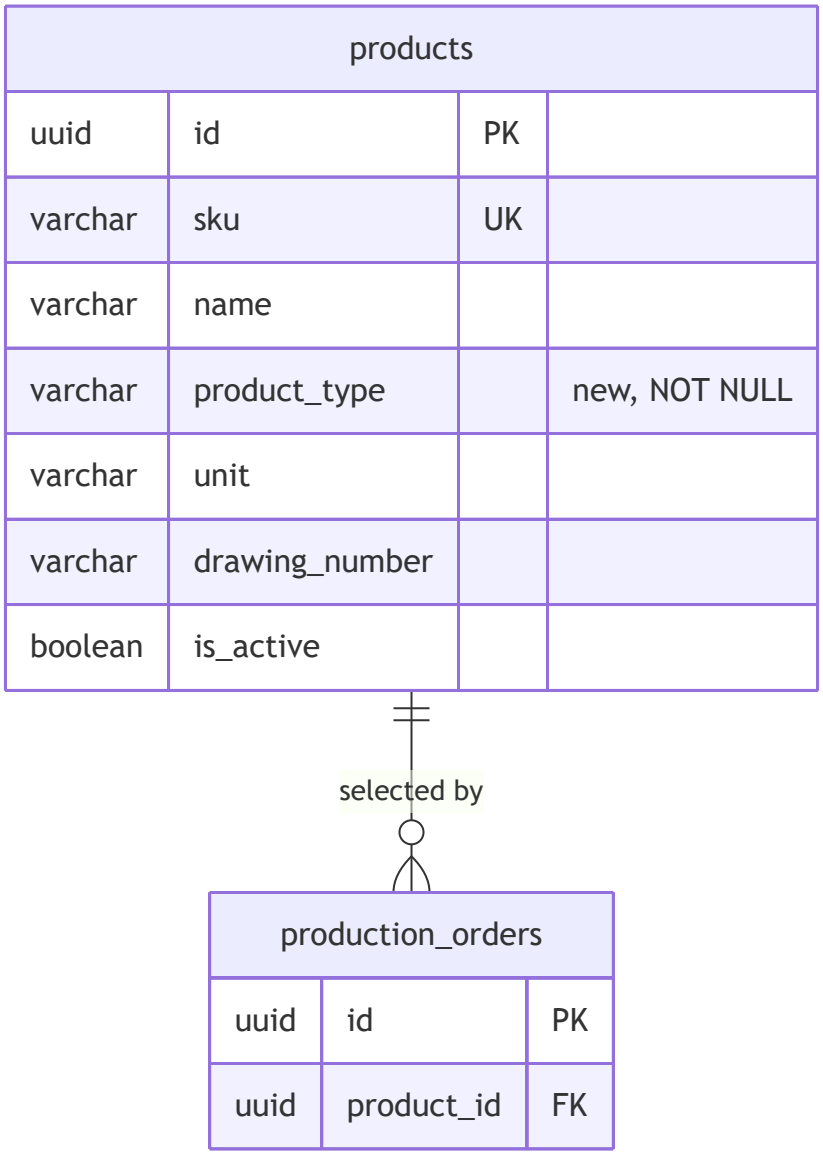

products
uuid
id
PK
varchar
sku
UK
varchar
name
varchar
product_type
new, NOT NULL
varchar
unit
varchar
drawing_number
boolean
is_active
selected by
production_orders
uuid
id
PK
uuid
product_id
FK

Table definition — `products` (added column only)

No.	Logical field	Physical name	PostgreSQL type	Nullable	Default	Key / validation	Notes
10	Product type	`product_type`	`varchar(20)`	no	none	`ck_products_product_type`	New. Code from 004_BD-TYPE §3; labels are UI only. Mutable on edit, including retired and order-referenced products

Why a code in `varchar` + CHECK, not a PostgreSQL `enum` or a lookup table: it matches the existing `unit` and `production_orders.status` columns (CHECK on text), maps to a C# enum stored as string, and a later value change is a one-line constraint migration. A lookup table was excluded by DEC-001 (fixed list, no master screen).

No column default after the migration: every write must name a type, so the API's required-field rule is also enforced by `NOT NULL` (REQ-090). The default is used only inside the migration's backfill.

Index definitions (インデックス一覧)

None added. The list filter (FN-045) adds `product_type = @type` to a query over a small, paged catalog (30 demo rows), and a four-value column has low selectivity; 004_DB's reasoning for not adding search indexes applies. Reassess with measured growth.

Trigger definitions

None.

Constraints

Constraint	Type	Rule
`ck_products_product_type`	CHECK	`product_type IN ('FINISHED', 'SEMI_FINISHED', 'PART', 'RAW_MATERIAL')`
`product_type NOT NULL`	NOT NULL	Enforced after backfill

Existing constraints (004_DB) unchanged.

Initial type mapping for the 30 demo parts (DEC-005)

Applied by SKU (immutable, 004_DB). Assemblies a customer receives are 完成品; machined or formed bodies that are further processed are 半製品; single components are 部品. None of the demo parts is a raw material, so 「原材料」 has no demo row; a tester creates one to see it.

SKU	Part	Type	SKU	Part	Type
P-1001	ブレーキキャリパー	FINISHED 完成品	P-1016	ボディワイヤーハーネス	FINISHED 完成品
P-1002	ブレーキディスクローター	PART 部品	P-1017	ヘッドランプユニット	FINISHED 完成品
P-1003	ブレーキパッド	PART 部品	P-1018	リレーボックス	PART 部品
P-1004	ドライブシャフト	FINISHED 完成品	P-1019	オルタネーター	FINISHED 完成品
P-1005	等速ジョイント	PART 部品	P-1020	ECU ケース	SEMI_FINISHED 半製品
P-1006	トランスミッションケース	SEMI_FINISHED 半製品	P-1021	インタークーラー	FINISHED 完成品
P-1007	ハブベアリング	PART 部品	P-1022	電動ファン	FINISHED 完成品
P-1008	ウォーターポンプ	FINISHED 完成品	P-1023	オイルフィルター	PART 部品
P-1009	エンジンマウント	PART 部品	P-1024	スロットルボディ	FINISHED 完成品
P-1010	ラジエーター	FINISHED 完成品	P-1025	ピストン	SEMI_FINISHED 半製品
P-1011	タイミングギア	PART 部品	P-1026	コネクティングロッド	SEMI_FINISHED 半製品
P-1012	クラッチディスク	PART 部品	P-1027	サブフレーム	FINISHED 完成品
P-1013	ショックアブソーバー	FINISHED 完成品	P-1028	ドアパネル	SEMI_FINISHED 半製品
P-1014	コイルスプリング	PART 部品	P-1029	ドアヒンジ	PART 部品
P-1015	エンジンワイヤーハーネス	FINISHED 完成品	P-1030	ボルト・ナットキット	PART 部品

Totals: 13 完成品, 5 半製品, 12 部品, 0 原材料. Any product not in this table (created through SCR-004 or by tests) gets `FINISHED`.

Database role and privileges

Role	Change	Reason
`pmai_app`	Column-scoped `UPDATE` becomes `(name, product_type, unit, drawing_number, is_active, updated_at_utc)` plus whatever the current grant already lists; `INSERT` / `SELECT` already cover new columns	Edit the type; `sku`, `id`, `created_at_utc` stay non-updatable

Role	Change	Reason
Migration owner	DDL and backfill	Unchanged

The exact current grant is read from the migrations during implementation and extended, not replaced.

API / DD mapping

Database field	API field	Design use
`product_type`	`type` (string code)	Product master list items, detail, create and update; optional `type` list filter. Not added to `GET /api/products` or order reads (DEC-003). Exact contract in 004_DD-TYPE

Migration impact and recovery

One EF Core migration, run by the owner before the new API is deployed (deploy/README.md):

1. `ALTER TABLE products ADD COLUMN product_type varchar(20) NOT NULL DEFAULT 'FINISHED'` — PostgreSQL 17 adds a constant default without rewriting the table, and every existing row reads `FINISHED`.
2. `UPDATE products SET product_type = … WHERE sku IN (…)` for the non-`FINISHED` demo rows in the table above (17 rows). Rows already edited by users keep their SKU, so the mapping still applies; absent SKUs are no-ops.
3. `ALTER COLUMN product_type DROP DEFAULT`; add `ck_products_product_type`.
4. Extend the `pmai_app` column `UPDATE` grant.

All four steps run in the migration's transaction, so a failure leaves the table as before. The old API keeps working between migration and deployment only for reads; its product create would fail `NOT NULL`, so migration and API deploy go together, as in earlier WIs. `Down` drops the constraint and column (type data is lost; nothing else depends on it). Verify upgrade from the current `master` database and a fresh database in revision 2.